(1) Funcția exponențială

$\exp z : \mathbb{C} \to \mathbb{C}^*$

$$\exp z = e^z = \sum_{n \geq 0} \frac{z^n}{n!}$$

Prop

1) $z^0 = 1$
2) $e^{z_1+z_2} = e^{z_1} \cdot e^{z_2} \quad \forall z_1, z_2 \in \mathbb{C}$
3) $e^{iy} = \cos y + i \sin y \quad \forall y \in \mathbb{R}$ (formula lui Euler)
4) funcția e^z este olomorfă și periodică de perioadă $2\pi i$

Demo

3) $e^{iy} = \sum_{n \geq 0} \frac{(iy)^n}{n!} = 1 + \frac{iy}{1!} - \frac{y^2}{2!} - \frac{iy^3}{3!} + \frac{y^4}{4!}$

$$e^{iy} = \underbrace{1 - \frac{y^2}{2!} + \frac{y^4}{4!} + \dots}_{\cos y} + i\underbrace{\left(\frac{y}{1!} - \frac{y^3}{3!} + \dots\right)}_{\sin y} = \cos y + i \sin y$$

4) $e^z = e^{x+iy} = e^x \cdot e^{iy} = e^x(\cos y + i \sin y) = \underbrace{e^x \cos y}_{P} + i\underbrace{e^x \sin y}_{Q}$

$P, Q \in C^1(\mathbb{R}^2)$

$\frac{\partial P}{\partial x} = e^x \cos y \qquad \frac{\partial P}{\partial y} = -e^x \sin y$

$\frac{\partial Q}{\partial y} = e^x \cos y \qquad \frac{\partial Q}{\partial x} = e^x \sin y$

$\Rightarrow \frac{\partial P}{\partial x} = -\frac{\partial Q}{\partial y}$ C-R $\xRightarrow{\text{Cor!}}$

$\Rightarrow e^z$ olomorf

$e^{z+2\pi i} = e^{x+iy+2\pi i} = e^x \cdot e^{i(y+2\pi)} = e^x(\cos(y+2\pi) + i \sin(y+2\pi))$

$= e^x(\cos y + i \sin y) = e^x \cdot e^{iy} = e^{x+iy} = e^z \Rightarrow e^z$ e periodică cu perioada $2\pi i$

(2) Fct. logaritmică

Se pune problema găsirii sol. $w \in \mathbb{C}$ ale ec. $e^w = z$, $z \in \mathbb{C}^*$ fixat

$w = u + iv$

$z = r(\cos\theta + i\sin\theta)$ $\quad r = |z|$

$\theta = \arg z \quad \theta \in [0, 2\pi]$

$e^{u+iv} = r(\cos\theta + i\sin\theta) = re^{i\theta}$

$e^u \cdot e^{iv} = re^{i\theta} \Rightarrow \begin{cases} e^u = r \Rightarrow u = \ln r = \ln|z| \\ v = \theta + 2k\pi = \arg z + 2k\pi \end{cases} \quad k \in \mathbb{Z}$

$$w = \ln z = \{\ln|z| + i(\arg z + 2k\pi), k \in \mathbb{Z}\}$$

(3) Fct. putere

$z^u = e^{u \ln z}$

$u \in \mathbb{C}$

$z^u = \{e^{u[\ln|z| + i(\arg z + 2k\pi)]}, k \in \mathbb{Z}\}$

$z^u = \{|z|^u \cdot e^{iu(\arg z + 2k\pi)}, k \in \mathbb{Z}\}$

(4) Fct. radical

$\sqrt[u]{z} = z^{\frac{1}{u}} = e^{\frac{1}{u}\ln z}$

$u \geq 2$

$\sqrt[u]{z} = \{e^{\frac{1}{u}[\ln|z| + i(\arg z + 2k\pi)]}, k \in \mathbb{Z}\}$

$\sqrt[u]{z} = \left\{\sqrt[u]{|z|} \cdot e^{i \cdot \frac{\arg z + 2k\pi}{u}}, k \in \mathbb{Z}\right\}$

$\sqrt[u]{z} = \left\{\sqrt[u]{|z|} \cdot e^{i \cdot \frac{\arg z + 2k\pi}{u}}, k \in \overline{0, u-1}\right\}$

ex $i^i = e^{i\ln i} = \{e^{i[\ln|i| + i(\arg i + 2k\pi)]}, k \in \mathbb{Z}\}$

$z = |z|(\cos\theta + i\sin\theta)$

$\frac{z}{|z|} = \cos\theta + i\sin\theta \qquad i = \cos\theta + i\sin\theta$

$$i^i = \{e^{-(\frac{\pi}{2} + 2k\pi)}, k \in \mathbb{Z}\}$$

ex $(1-i)^{1+i} = e^{(1+i)\ln(1-i)}$

$$\ln(1-i) = \{\ln|1-i| + i(\arg(1-i) + 2k\pi), k \in \mathbb{Z}\}$$

$$\frac{1-i}{\sqrt{2}} = \cos\theta + i\sin\theta \Rightarrow \frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2} = \cos\theta + i\sin\theta$$

$$\theta = 2\pi - \frac{\pi}{4} = \frac{7\pi}{4}$$

$$\ln(1-i) = \{\ln\sqrt{2} + i(\frac{7\pi}{4} + 2k\pi), k \in \mathbb{Z}\}$$

$$(1-i)^{1+i} = \{e^{(1+i)[\ln\sqrt{2} + i(7\pi/4 + 2k\pi)]}, k \in \mathbb{Z}\}$$

$$= \{e^{\ln\sqrt{2} - (7\pi/4 + 2k\pi)} \cdot e^{i(\ln\sqrt{2} + 7\pi/4 + 2k\pi)}, k \in \mathbb{Z}\}$$

$$(1-i)^{1+i} = \{\sqrt{2}e^{-7\pi/4 - 2k\pi}[\cos(\ln\sqrt{2} + 7\pi/4 + 2k\pi) + i\sin(\ln\sqrt{2} + 7\pi/4 + 2k\pi)], k \in \mathbb{Z}\}$$

$$(1-i)^{1+i} = \{\sqrt{2}e^{-7\pi/4 - 2k\pi}[\cos(\ln\sqrt{2} + 7\pi/4) + i\sin(\ln\sqrt{2} + 7\pi/4)], k \in \mathbb{Z}\}$$

3) Să se rezolve următoarea ecuație $z^8 + 2z^4 + 2 = 0$

$z^4 = t \Rightarrow t^2 + 2t + 2 = 0$

$$t_{1,2} = -1 \pm i$$

$$z^4 = -1 + i \Rightarrow z_k = \sqrt[4]{-1+i} = (-1+i)^{1/4} = e^{1/4\ln(-1+i)}$$

$$z_k = \left\{ e^{1/4\left[\ln|-1+i| + i(\arg(-1+i)+2k\pi)\right]},\ k \in \overline{0,3} \right\}$$

$$\frac{-1+i}{\sqrt{2}} = \cos\theta + i\sin\theta = -\frac{\sqrt{2}}{2} + i\cdot\frac{\sqrt{2}}{2} \Rightarrow \theta = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$$

$$z_k = \left\{ e^{1/4\left[\ln\sqrt{2} + i(3\pi/4 + 2k\pi)\right]},\ k \in \overline{0,3} \right\}$$

$$z^4 = -1-i \Rightarrow z_k = \sqrt[4]{-1-i} = e^{1/4\ln(-1-i)}$$

$$z_k = \left\{ e^{1/4\left[\ln|-1-i| + i(\arg(-1-i)+2k\pi)\right]},\ k = \overline{0,3} \right\}$$

$$\frac{-1-i}{\sqrt{2}} = \frac{-\sqrt{2}}{2} - i\frac{\sqrt{2}}{2} \Rightarrow \theta = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$$

$$z_k = \left\{ e^{1/4\left[\ln\sqrt{2} + i(5\pi/4 + 2k\pi)\right]},\ k = \overline{0,3} \right\}$$

(5) Fct. trigonometrice

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i} \qquad \cos z = \frac{e^{iz} + e^{-iz}}{2}$$

$$\operatorname{sh} z = \frac{e^{z} - e^{-z}}{2} \qquad \operatorname{ch} z = \frac{e^{z} + e^{-z}}{2}$$

ex $\operatorname{tg} z = \frac{4-3i}{5} \Rightarrow \frac{\sin z}{\cos z} = \frac{4-3i}{5} \Rightarrow \frac{e^{iz} - e^{-iz}}{2i} \cdot \frac{2}{e^{iz} + e^{-iz}} = \frac{4-3i}{5}$

$$5e^{iz} - 5e^{-iz} = 4ie^{iz} + 4ie^{-iz} + 3e^{iz} + 3e^{-iz} \ \Big|\cdot e^{iz} \Rightarrow$$

$$\Rightarrow 5e^{2iz} - 5 = 4ie^{2iz} + 4i + 3e^{2iz} + 3 \Rightarrow$$

$$\Rightarrow e^{2iz}(2-4i) = 4i + 8 \Rightarrow e^{2iz} = \frac{4i+8}{2-4i} = \frac{2i+4}{1-2i}$$

$$e^{2iz} = \frac{(2i+4)(1+2i)}{5} = \frac{2i+4-4+8i}{5} = \frac{10i}{5} = 2i \Rightarrow$$

$$\Rightarrow 2iz = \ln(2i)$$

$$\Rightarrow z = \frac{1}{2i}\ln(2i) = \left\{-\frac{i}{2}\left[\ln|2i| + i(\arg(2i) + 2k\pi)\right], k \in \mathbb{Z}\right\}$$

$$= \left\{-\frac{i}{2}\ln 2 + \frac{1}{2}\left(\frac{\pi}{2} + 2k\pi\right), k \in \mathbb{Z}\right\}$$